

# Nick DiSanto

<https://nickdisanto.github.io>

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## Education

### Vanderbilt University

Jan. 2025 – Present

PhD in Computer Science

GPA: 4.0/4.0. Advised by Dr. Ipek Oguz.

### California Baptist University

Aug. 2019 – Apr. 2023

B.S. in Computer Science, *summa cum laude*

GPA: 4.0/4.0. Ranked #1 in graduating class.

## Experience

### Vanderbilt Medical Image Computing Lab

Jan. 2025 – Present

*Graduate Student Researcher*

- Developing deep learning methods for image reconstruction and restoration
- Designing registration and volumetric inpainting models for high-resolution 3D OCT data, addressing motion-induced slice corruption

### GE Vernova

Jan. 2024 – Jan. 2025

*Software Engineer*

- Led the wildfire forecasting initiative, including the end-to-end training, visualization, and productization of ML pipelines with power grid applications
- Presented results of wildfire risk forecasting analytic to the C-Suite, helping to secure significant funding for further integration
- Honored as a 2024 GE Vernova Changemaker for co-leading a company-wide initiative to integrate GenAI into GEV Electrification Software products

### CalBaptist ML & NLP Lab

Jan. 2022 – Dec. 2023

*Lead Undergraduate Researcher*

- Built representation models to find empirical patterns in unstructured online data, challenging the necessity of complex algorithms for natural language pattern recognition
- Proposed a novel data preprocessing approach that introduces noise and helps medical imaging models better perceive their environments, increasing accessibility for low-income communities
- Produced analyses and applications for three papers, leading the team from inception to execution

### Sirch

Sep. 2023 – Dec. 2023

*NLP Consultant*

- Built models that perform Query Auto-Completion (QAC), information retrieval, and NLU
- Scaled product to efficiently handle increasing foot traffic while maintaining a personalized UI

### General Electric

June 2022 – Aug. 2023

*Software Engineering Intern*

- Led a database optimization initiative that saves thousands of dollars per year
- Developed and optimized microservices for large-scale data ingestion applications

### Keysight Technologies

June 2021 – Aug. 2021

*Software Engineering Intern*

- Created a web platform and built applications to enable clients to customize their products
- Performed data analysis to help eliminate 40% of product-option configurations

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|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Publications & Preprints | <hr/>                                                                                                                                                                                                                                                                                                                                        |                   |
|                          | <i>LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching.</i><br>X Yao, R Yu, <b>N DiSanto</b> , E Aghdam, K Oguine, D Lu, A Lou, J Wang, D Hu, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz. <i>Medical Imaging with Deep Learning (MIDL)</i> , 2025. <a href="#">[pdf]</a> <a href="#">[code]</a> |                   |
|                          | <i>Spatial Analysis of Social Media’s Proxies for Human Emotion and Cognition.</i><br>A Corso, <b>N DiSanto</b> , N Corso, E Lee. <i>International Conference on Information</i> , 2024.                                                                                                                                                     |                   |
|                          | <i>Transcending the Attention Paradigm: Representation Learning from Geospatial Social Media Data.</i><br><b>N DiSanto</b> , A Corso, B Sanders, G Harding. <i>arXiv preprint arXiv:2310.05378</i> , 2023. <a href="#">[pdf]</a> <a href="#">[code]</a>                                                                                      |                   |
|                          | <i>Leveraging Contextual Data Augmentation for Generalizable Melanoma Detection.</i><br><b>N DiSanto</b> , G Harding, E Martinez, B Sanders. <i>arXiv preprint arXiv:2212.05116</i> , 2022. <a href="#">[pdf]</a> <a href="#">[code]</a>                                                                                                     |                   |
| Teaching Experience      | <i>Beyond Interpretable Benchmarks: Contextual Learning through Cognitive and Multimodal Perception.</i><br><b>N DiSanto</b> . <i>arXiv preprint arXiv:2304.00002</i> , 2022. <a href="#">[pdf]</a>                                                                                                                                          |                   |
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|                          | EGR121: <b>Intro to C++</b> , <i>Teaching Assistant</i>                                                                                                                                                                                                                                                                                      | SP 2023           |
|                          | EGR225: <b>Discrete Structures</b> , <i>Teaching Assistant</i>                                                                                                                                                                                                                                                                               | FA 2022           |
|                          | CSC312: <b>Algorithms</b> , <i>Tutor</i>                                                                                                                                                                                                                                                                                                     | SP/FA 2022        |
|                          | EGR329: <b>Computer Architecture</b> , <i>Tutor</i>                                                                                                                                                                                                                                                                                          | SP/FA 2022        |
| Awards & Honors          | PHY201: <b>Physics for Engineers</b> , <i>Teaching Assistant</i>                                                                                                                                                                                                                                                                             | SP/FA 2021        |
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|                          | 2024 GE Vernova Changemaker (selected out of 1200 nominees)                                                                                                                                                                                                                                                                                  | Nov. 2024         |
|                          | CS Outstanding Student Award (ranked #1/40)                                                                                                                                                                                                                                                                                                  | Apr. 2023         |
|                          | Inducted into Alpha Chi Honor Society (top 10%)                                                                                                                                                                                                                                                                                              | Apr. 2022         |
|                          | Physics Performance & Leadership Award                                                                                                                                                                                                                                                                                                       | Dec. 2021         |
|                          | President’s List (every semester of undergraduate study)                                                                                                                                                                                                                                                                                     | FA 2019 – SP 2023 |
|                          | Trustee Merit Scholarship                                                                                                                                                                                                                                                                                                                    | FA 2019 – SP 2023 |
| Leadership & Outreach    | Engineering Excellence Scholarship                                                                                                                                                                                                                                                                                                           | SP 2021 - SP 2023 |
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|                          | Founder and President of CBU Chess Club.                                                                                                                                                                                                                                                                                                     |                   |
|                          | Member of CBU’s Association for Computing Machinery (ACM) Chapter.                                                                                                                                                                                                                                                                           |                   |
|                          | Lead trumpet and section leader in 4 competitive CBU musical ensembles.                                                                                                                                                                                                                                                                      |                   |
|                          | Set up a newly-opened local high school’s computer lab.                                                                                                                                                                                                                                                                                      |                   |
|                          | Performed STAR test data analysis for an underresourced elementary school during COVID-19 pandemic.                                                                                                                                                                                                                                          |                   |